

Analytical II

Lec 1

Quantitative analysis

According to Sample Conc.

Macro-analysis

> 100 mg Sample

Semi-micro analysis

10:100 mg Sample

Micro-analysis

< 1 mg Sample

According to analysis method

Gravimetric

By weight

Volumetric

تركيب
معروف

Standard ≠ Sample

Complete Reaction

غير
معروف

Instrumental

Based on optical
& electrical
phenomena.

Volumetric Analysis

- Adding titrant to solution contains titrand and let them to react. $\rightarrow$ known conc. $\rightarrow$ unknown

Types of Volumetric Analysis

- ① Acid-Base: Acidic or basic titrant reacts with acid or base titrand $\rightarrow$ form ionized water.
- ② Complexometric: Based on metal-ligand complexation.
- ③ Precipitation: Titrant + Titrand $\rightarrow$ Precipitate
- ④ Redox: The titrant is an oxidizing or reducing agent.

Requirements for titrimetric reaction

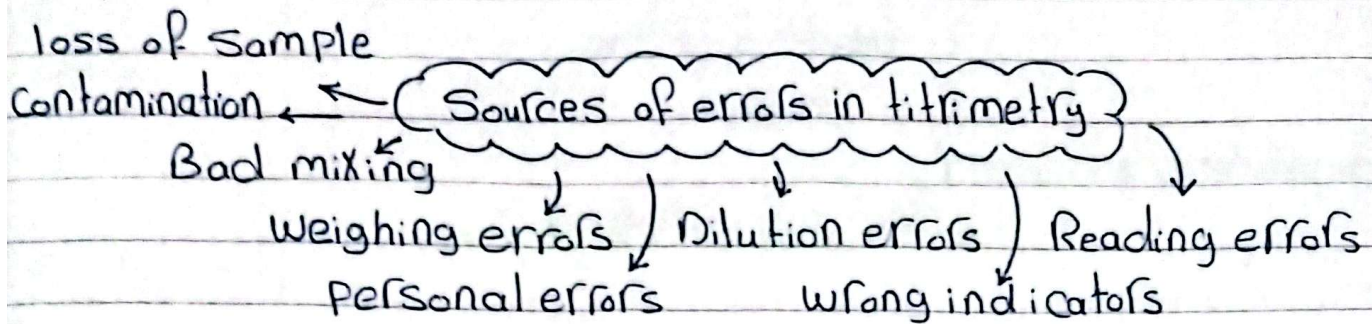
No Side Reaction

Very Rapid

Simple Reaction

Easy to detection the end point

Available Suitable
Standard Solution



standard Solution

Molar (M)

Contain gram M.wt in 1 liter.
Known Conc.

Normal (N)

Contain gram Eq. wt
in 1 liter.

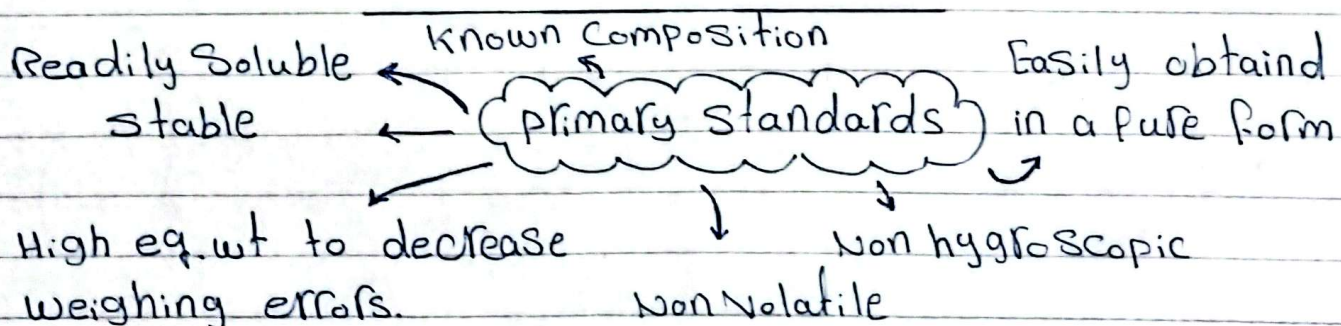
① Equivalent weight of Acids = $\frac{\text{Molecular weight}}{\text{number of replaceable } H^+}$

② " " Bases = $\frac{\text{Molecular weight}}{\text{number of replaceable } OH^-}$

③ " " Salts = $\frac{\text{Molecular weight}}{\text{Number of ions} \times \text{Valency}}$ → السالتي

* Solutions have the same normalities react together in equal volumes.

يعني لو حطين و قلا عدة بنفس النورماليتي اتفاعلوا مع بعض كل واحد منهم بنفس الكمية هيتعمل تعادل كامل " مفيش مرفا زيادة عن التاني "



- Secondary standards → primary
↳ Require standardization against a primary standard.

preparation of standard solutions

Direct method

Indirect method

Solute must be of primary standard quality.

Is standardization against a primary standard

Acid-Base Titrations

Dissociation Theory

Molecules $\rightleftharpoons$ Cations + Anions
"electrolyte"

Degree of dissociation (α)

$\alpha = \frac{\text{Number of solute molecule dissociation}}{\text{Total number of solute molecule before dissociation}}$

α near to one

Complete dissociation

Strong electrolyte

Far from one

Weak dissociation

Weak electrolyte

Acid-Base Theories

Arrhenius Theory

Acid

Base

Ionizes to give H^+

Ionizes to give OH^-

→ Didn't discuss the role of solvent in the ionisation.

Brønsted-Lowry Theory

↓
Acid

Donates H^+

↓
Base

Accepts H^+

The Stronger the acid, the weaker its conjugate base.

↓ علاقة عكسية بين قوة الحمض والقاعدة

water is amphoteric → Act as an acid or a base.

Law of mass action

→ Rate of Forward Reaction $v_f \propto [A]^a [B]^b$

$$v_f = k_f [A]^a [B]^b$$

→ Rate of Backward Reaction $v_b \propto [C]^c [D]^d$

$$v_b = k_b [C]^c [D]^d$$

↓ حالة الاتزان

$$v_f = v_b$$

$$\rightarrow k_f [A] \cdot [B] = k_b [C] \cdot [D]$$

$$\rightarrow \frac{k_f}{k_b} = \frac{[C] \cdot [D]}{[A] \cdot [B]}$$

$$\rightarrow k_f / k_b = K_{eq}$$

$$\rightarrow K_{eq} = \frac{[C] \cdot [D]}{[A] \cdot [B]}$$

Lewis Theory

↓
Acid

Accepts electrons

↓
Base

Donates electrons

<< Dissociation of Water >>

• Water ionization equation: $H_2O \rightleftharpoons H^+ + OH^-$

↳ ionization constant: $K_c = \frac{[H^+][OH^-]}{[H_2O]}$

• in dil. solutions: $K_c[H_2O] = [H^+][OH^-]$ ion product of water

$K_w = [H^+][OH^-] = 1 \times 10^{-14}$ dissociation constant

• In pure water: $[H^+] = [OH^-] = 1 \times 10^{-7}$

• $pH = -\log[H^+]$ $pOH = -\log[OH^-]$ $pK_w = pH + pOH = 14$

$pH = pK_w - pOH$

$pOH = pK_w - pH$

$pH = pOH \rightarrow$ in neutral solutions

$pH < 7 \rightarrow$ in acidic solutions

$pH > 7 \rightarrow$ in alkaline solutions

* imp. of pH measurements: 1- Formulation of aq. solutions

2- absorption, distribution, excretion of drug

3- pharmaceutical analysis

<< pH of acids, bases, solutions >>

strong acid & bases are completely dissociated

∴ concentration = $[H^+]$ or $[OH^-]$

Weak acids & bases only small amount is dissociated

$pH = \frac{1}{2} pK_a + \frac{1}{2} pC_a \rightarrow$ pH of weak $\bar{a}$

$pH = 14 - \frac{1}{2} pK_b - \frac{1}{2} pC_b \rightarrow$ pH of weak B^-

pH of salts of salt solutions

Salt of strong $\bar{a}$ & B^-

$pH = 7$

Salt of strong $\bar{a}$ & weak B^-

$pH < 7$

$pH = \frac{1}{2} pK_w - \frac{1}{2} pK_b + \frac{1}{2} pC_s$

Salt of weak $\bar{a}$ & strong B^-

$pH > 7$

$pH = \frac{1}{2} pK_w + \frac{1}{2} pK_a - \frac{1}{2} pC_s$

Salt of weak $\bar{a}$ & B^-

$pH = \frac{1}{2} pK_w + \frac{1}{2} pK_a - \frac{1}{2} pK_b$

« Buffer solutions »

Solutions that resist the change of pH upon the addition of small amount of acids or bases.

Types of buffers

Acidic buffers $\text{CH}_3\text{COOH}, \text{CH}_3\text{COONa}$ Consist of weak A & its salt	Basic buffer $\text{NH}_4\text{OH}, \text{NH}_4\text{Cl}$ Consist of weak B & its salt
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Importance → 1- control pH of liq. formulations

→ 2- Optimize stability, solubility of dissolved drug

→ 3- pH of blood is very well buffered

Henderson Basel equation for calc. pH of buffers :

pH of acidic buffers :

$$\text{pKa} = \text{pH} - \log \frac{[\text{Salt}]}{[\text{acid}]}$$

$$\text{pH} = \text{pKa} + \log \frac{[\text{Salt}]}{[\text{acid}]}$$

pH of basic buffers :

$$\text{pH} = \text{pK}_w - \text{pK}_b - \log \frac{[\text{Salt}]}{[\text{base}]}$$

« Buffer capacity »

The limit to how much acid or base can be added to a given buffer solution before any appreciable change in pH.

Buffer capacity range → 1:10 or 10:1

• **Buffer ratio** : is the ratio $\frac{[\text{Salt}]}{[\text{acid}]}$

$$\text{pH} = \text{pKa} \pm 1$$

• **Max. buffer capacity** : when $[\text{Salt}] = [\text{acid}]$ (pH = pKa)

« Indicators »

They are used for detecting end point in volumetric analy.

Types of indicators

color indicators	Flourescent indicators	Turbidity indicators
Organic Substances with 1 M.Wt →	Organic compd. emit fluorescence at certain pH →	Weak organic compd. forming colloidal ppt →
1 color: phenolphthalein	Fluorescence	iso-nitroacetyl
2 color: methyl orange		p- amino azo-benzene

End point in acid-base titration can be detected by using neutralization indicators of acid-base indicators

Analytical II

Lec 8

Titration Curves

↳ The plot of pH & Volume of titrant.

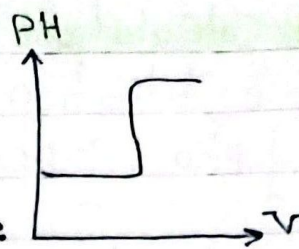
- we have to calculate the pH at 4 stages:

① Before titration.

② During titration.

③ At the equivalence point.

④ After the equivalence point.



Calculating the pH

→ Mathematically: By equations.

or → practically: By pH meter.

Types of titration Curves

1. **Strong Acid-Strong Base** (HCl - NaOH)

→ pH Calculating at diff. stages:-

- Before Titration

$$[H^+] = 0.1 \rightarrow pH = -\log(0.1) = 1$$

- During Titration (+ 10 ml NaOH)

$$pH = -\log \frac{(0.1 \times 15)}{35} = 1.377$$

- At the equivalence point (+ 25 ml NaOH)

$$pH = 7$$

- After the equivalence point (+ 26 ml NaOH)

$$pH = 14 - (-\log 0.00196) = 11.29$$

→ Indicator selection:-

↳ Any indicator with a color transition at pH (3.5 - 10.5)

ex. → Phenolphthalein, Phenol Red, Methyl orange, Methyl Red.

* Alizarin yellow → isn't suitable (out of range)

Rises very slowly until near end point then suddenly rises to pH = 7 then to pH = 10

2. Weak Acid - Strong Base ($\text{CH}_3\text{COOH} - \text{NaOH}$)

→ pH Calculation at diff. stages:-

- Before titration

$$\text{pH} = \frac{1}{2} \text{pK}_a + \frac{1}{2} \text{pC}_a = \frac{1}{2} (4.74) + \frac{1}{2} (-\log 0.1) = 2.94$$

- During titration (+ 10 ml NaOH)

$$\text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]} = 4.56$$

(At Equivalence point (+ 25 ml NaOH))

$$\text{pH} = 9.37$$

(After Equivalence point (+ 35 ml NaOH))

$$\text{pH} = 12.22 \rightsquigarrow (14 - 1.77)$$

→ Indicator Selection:-

- phenolphthalein → is the only suitable indicator.

3. Weak Base - Strong Acid ($\text{NH}_4\text{OH} - \text{HCl}$)

→ pH Calculating at diff. stages:-

- Before Titrations

$$\text{pH} = 14 - \frac{1}{2} \text{pK}_b - \frac{1}{2} \text{pC}_b = 11.13$$

- During titration (+ 10 ml HCl)

$$\text{pH} = 9.44$$

- At Equivalence point (+ 25 ml HCl)

$$\text{pH} = 5.28$$

- After Equivalence point (+ 35 ml HCl)

$$\text{pH} = -\log [\text{H}^+] = 1.77$$

→ Indicator Selection

e.x → Methyl orange, Methyl Red → Are the suitable indicators

4. Weak Acid - Strong Base

- pH change near the equivalence point is gradual.
- No single indicator is effective.

→ Detection methods :-

① pH meter "Electrochemical method"

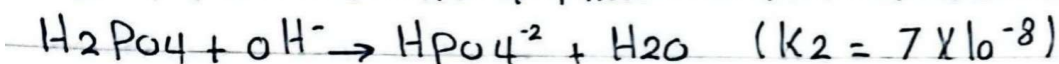
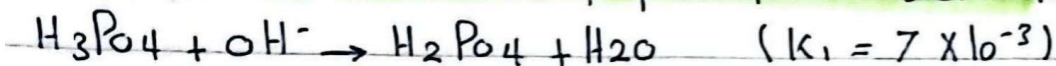
② Mixed Indicators

e.x → Neutral Red + Methylene blue

↳ For acetic acid & ammonium hydroxide

5. Polybasic Acid - Strong Base (H_3PO_4 - NaOH)

→ Stepwise Reactions of phosphoric acid (H_3PO_4) with OH^- :



* The third proton ($pK_3 = 12.32$) is weakly dissociated

Case	pH Equation	Indicator	Acid color	pH range	Base color
Strong acid	$pH = -\log[H^+]$	Alizarin Yellow R		10.1-12.0	
Strong base	$pH = 14 - (-\log[H^+])$	Bromocresol green		3.8-5.4	
Weak acid	$pH = \frac{1}{2} pK_a + \frac{1}{2} pC_a$	Bromocresol purple		5.2-6.7	
Weak base	$pH = pK_w - \frac{1}{2} pK_b - \frac{1}{2} pC_b$	Bromophenol blue		3.0-4.6	
Acidic buffer	$pH = pK_a + \log [\text{salt}]/[\text{acid}]$	Bromothymol blue		6.0-7.6	
Basic buffer	$pH = pK_w - pK_b - \log [\text{salt}]/[\text{base}]$	Litmus		4.5-8.3	
Salt of weak acid & strong base	$pH = \frac{1}{2} pK_w + \frac{1}{2} pK_a - \frac{1}{2} pC_s$	Malachite green		0.2-1.8	
Salt of weak base & strong acid	$pH = \frac{1}{2} pK_w - \frac{1}{2} pK_b + \frac{1}{2} pC_s$	Methyl orange		3.1-4.4	
		Methyl red		4.2-6.3	
		Methyl violet		0.0-1.6	
		Phenol red		6.6-8.0	
		Phenolphthalein		8.2-10.0	

نعرف الحاجات المشهورة بس

Analytical II

Lec 4

Neutralization titrations

Direct

- of Acid & Bases

Double indicator

- Mix. of HCl

& CH_3COOH

- H_3PO_4

Easily Hydrolyzable Salts

→ Salt of strong Base & very weak acid: - KCN

- Na_2CO_3 - Borax

Applications

→ Determination of Borax.

→ " mixture of Borax & Boric acid.

Direct titration of acids & bases

1. Acid Determination

→ Strong acids : e.x (HCl - H_2SO_4 - HNO_3)

- Titrated against NaOH using ph.ph or methyl orange.

→ Weak acids : e.x (HCOOH - CH_3COOH)

- Titrated against NaOH using ph.ph only.

→ Water-insoluble acids : e.x (Benzoic acid)

- Dissolved in neutral ethanol before titration with NaOH using ph.ph.

→ Acidic pharmaceutical compounds : e.x (aspartic A - Naproxen)

- Titrated with a standard base.

→ Boric acid Titration (very weak acid) $K_a = 5.8 \times 10^{-10}$

- To enhance acidity → + poly hydroxy compound (Glycerol)

- Glycorborate is titrated with NaOH using ph.ph

Selecting an appropriate indicator based on the reactant

2. Base Determination

- **Strong bases**: e.x (NaOH - KOH - Ba(OH)₂)
- Titrated with standard acids using ph.ph or methyl orange
- **weak bases**: Titrated with HCl using Methyl Red or Methyl orange.
- **pharmaceutical bases**: e.x (Aminophylline)
- Titrated with HCl

Double indicator Titrations

- Detect multiple equivalence points in:
 - Dibasic acids (H₂SO₄ - H₂S - H₂CO₃)
 - Polybasic acids (Phosphoric acid - H₃PO₄)
 - Mix. of two monobasic acids (HCl & CH₃COOH)

Requirements for its applications :-

1. Ionization constant (K_a) must be $\geq 10^{-7}$
2. K_a Ratio between strong & weak acids must be $\geq 10^4$ to ensure distinct inflection points in the titration curve.

Titration of Diprotic Acids

- 1st Equivalence Point →
- using Methyl orange.
- 2nd Equivalence Point →
- Using ph.ph

$$pH_1 = \frac{pK_1 + pK_2}{2}$$

$$pH_2 = \frac{1}{2} pK_w + \frac{1}{2} pK_a - \frac{1}{2} pC_s$$

→ Titration of poly-protic acid (H_3PO_4)

1. $H_3PO_4 \rightleftharpoons H^+ + H_2PO_4^-$ $pK_1 = 2.1$ by methyl orange
2. $H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-}$ $pK_2 = 7.2$ by ph.ph
3. $HPO_4^{2-} \rightleftharpoons H^+ + PO_4^{3-}$ $pK_3 = 12.7$ known directly
Because K_{a3} is too weak ←

→ Alternative method:-

Adding $CaCl_2$ to precipitate phosphate radicals, releasing an equivalent amount of HCl , which is then titrated.

→ Titration of a mixture of two Monobasic Acids (HCl & CH_3COOH)

- $NaOH$ first neutralizes HCl , followed by CH_3COOH .
- Common ion effect suppresses CH_3COOH ionization, ensuring HCl is neutralized first.
- Second equivalence point is in a basic range, require ph.ph

→ Titration of Easily hydrolyzable salts

- Hydrolyzable salts arise from:
 - ↳ strong base & very weak acid (KCN - Borax - Na_2CO_3)
 - ↳ strong acid & very weak base (aniline HCl - hydroxylamine HCl)

→ Titration of potassium Cyanide (KCN)



- The resulting KOH is titrated with HCl .

→ Titration of Borax

- Borax hydrolyzes, forming $NaOH$ & Boric acid.

Determination by

- Titration of liberated NaOH with 0.1 N HCl (V_1)
- Methyl orange indicator.

- Addition of Glycerol to enhance acidity, followed by titration with NaOH. (V_2)
- Ph. ph indicator

$$V_2 = 2V_1$$

→ Determination of Borax-Boric Acid mixture

- ① 1st titration with HCl (V_1) → Determines NaOH from Borax.
- ② 2nd titration with NaOH (V_2) → after Glycerol addition → Determines total Boric acid
- ③ Boric acid in the mixture = $V_2 - 2V_1$

• Advanced Acid-Base titrations

→ Double indicator titration for Na_2CO_3

- 1st endpoint (pH = 8.3 - ph.ph)
 $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow \text{H}_2\text{CO}_3 + 2\text{NaCl}$
- 2nd endpoint (pH = 3.8 - Methyl orange)
 $\text{NaHCO}_3 + \text{HCl} \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{NaCl}$ (complete neutralization)

→ Determination of Na_2CO_3 & NaHCO_3 mixture

① Same Sample

- $V_1 = \frac{1}{2}$ Carbonate (ph.ph)
- $V_2 = \text{Total Carbonate} + \text{bicarbonate}$ (Methyl orange)
- Bicarbonate = $V_2 - 2V_1$

② Different Samples

- 1st sample titrated with ph.ph ($V_1 = \frac{1}{2}$ Carbonate)
- Another sample titrated with methyl orange.
($V_2 = \text{Total Carbonate} + \text{bicarbonate}$)

Biphasic titration

• Titration of Water-soluble salt the $\bar{a}$ of which is insoluble in water but soluble in an immiscible organic solvent.

→ e.x Sodium Salicylate & Sodium benzoate

Titration of Sodium Salicylate

• Sample: Sod. Salicylate . Standard: HCl . Indicator: Bromophenol blue



Salicylic: → changes indicator's colour from basic (blue) to acidic (pale green) → False endpoint is obtained → Salysilic must be removed from aq. layer by shaking with ether.

Residual (back) titration

• Used when direct titration isn't suitable for:

- 1- Volatile Sub., e.x (Ammonia, Formic $\bar{a}$),
- 2- Insoluble Sub., e.x (ZnO, CaO, BaCO₃)
- 3- Sub. require heating with standard

Depend on known XSS. Standard reacts with sample, then residual amount of standard is determined.



→ boil until no more ammonia is evolved.

→ Residual NaOH is cooled & back titrated with standard HCl

• **Blank determination**: Separate determination in which all conditions are identical to the assay analysis, except the sample (Without the sample)

Blank determination is done to determine any titration error used for:

- 1- Standard changes its strength during assay.
- 2- Heating & Cooling during residual titration change in strength of standard alkali.
- 3- To determine exact volume of XSS. standard solution.
- 4- To minimize titration error.

* Application of residual titration:

Determination of: 1- ammonium salt

2- ^{insoluble} ZnO 3- ^{insoluble} CaCO₃ 4- ^{soluble} BaCl₂ 5- Salicylic acid (aspirin)
 6- mixture of [CaO + CaCO₃]: esters → acetylsalicylic acid

* Application of indirect titration:

1- ammonium salt 2- HgO 3- aldehydes 4- ketones

Determination of ammonium salts (NH₄Cl)

1- Residual titration method



• V of titrant equivalent to ammonia = 25 - V

2- Indirect titration method (Formal method)

→ Reaction of formaldehyde with ammonium chloride



Determination of insoluble Zinc oxide



Residual standard HCl is back titrated with standard NaOH using M.O or ph.ph
 $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

→ V. of standard NaOH - residual standard HCl = V

→ V. " " HCl equivalent to ZnO sample = 25 - V

Determination of insoluble CaCO₃



boil to remove CO₂ then cooled & residual HCl is back titrated against NaOH using M.O.

- blank experiment should be carried out

Determination of mixture of CaCO_3 & CaO



- mixture is boiled to remove $\text{CO}_2 \rightarrow$ residual HCl back titrated with NaOH using M.O

Blank experiment must be carried out

- CaO soluble in sucrose more than in water
(CaO & Ca saccharate $\rightarrow$ alkaline to ph.ph)

$\rightarrow$ mixture is suspended in alcohol + sucrose $\rightarrow$ soluble Ca saccharate $\rightarrow$ mixture titrated $\bar{e}$ HCl using ph.ph

Determination of soluble salts (BaCl_2 , CaCl_2 , SrCl_2)

$\rightarrow$ all are neutral salts

$\rightarrow$ BaCl_2 is soluble salt of neutral pH values, so it can't be determined by direct titration.

Determination of barium chloride

- 1- BaCl_2 + known xss. of $\text{Na}_2\text{CO}_3 \rightarrow \text{BaCO}_3$ (white ppt)
- 2- Boil solution, cooled $\rightarrow$ residual Na_2CO_3 is back titrated with standard HCl using ph.ph

Blank experiment must be carried out

Determination of ester (acetyl salicylic acid $\rightarrow$ aspirin)



The residual NaOH is back titrated $\bar{e}$ standard $\text{HCl} \rightarrow \text{ph.ph}$

Blank experiment must be carried out